DO TRONG KHANH

Software Engineer

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Ho Chi Minh City, Vietnam

Software engineer with 5+ years of experience building backend-heavy products: Kotlin and Spring Boot services, Go and Spark data pipelines, NestJS platforms and React frontends. I take features from spec to production (design, build, deploy and monitoring), with a focus on data integrity, resilient integrations and tests that cover failure modes, not just the happy path.

Experience

Spartan | Software Engineer **Mar 2025 – Present**

Software Engineer in US client product teams

LoanBud – small-business financing platform Jan 2026 – Oct 2026

Kotlin, Micronaut, Java, Spring Boot, PostgreSQL, Redis, AWS SQS, React, TypeScript, Terraform

- Rebuilt the Hub-to-CRM data sync on a **transactional outbox** (Postgres triggers, `SKIP LOCKED` claims, a horizontally scaled sync worker, version-guarded upserts), replacing a polling job that silently dropped failed syncs.
- Built an idempotent **Meta Lead Ads** pipeline (webhook → SQS → Graph API) and a schema-driven CRM segment builder over JSONB that fails closed; added marketing attribution and a per-field audit log.
- Shipped **Twilio video meetings** with calendar sync, hardened with a circuit breaker, room quotas and Redis-backed distributed rate limiting.
- Owned a rewrite of the legacy PHP core service in **Kotlin** (∼326 commits, +153k lines).

Dealops – CPQ software for complex pricing Jul 2025 – Jan 2026

React, TypeScript, tRPC, Prisma, Ant Design, Tailwind CSS, Turborepo

- **AI features**: built an in-app **AI chat assistant** from scratch (chat widget, its own AI chat client, streaming responses, markdown rendering), and shipped fixes and segment-based product filtering for the AI pricing assistant.
- **Admin Dashboard V2**: set up the new admin layout and built the product catalog (add/edit forms, clone and clone-and-save, tiered-pricing charts in drawers), product discount rules, and a terms library with segment filtering.
- **Approval rules**: refactored term approval rules to support direct-entry and selection types, with currency, percentage, date and tierable rule editors, a normalised tier setup and an **additional-logic builder** for compound conditions; made product selection respect affinity rules.
- **Opportunity page V2**: quote groups, create and clone quote flows, a quote proposals page with archive and restore, the submit-for-approval flow with approval nudges, activity history and recent opportunities.
- **Pricing quotes**: the **IC++ pricing model** with 3-decimal payment-method fees, ramp revenue logic and the ramp editor, side-by-side quote comparison, and single-quote slide generation with editable details.

Deep Sky Climate – IoT data platform for a carbon removal project developer Mar 2025 – Jul 2025

Go, Scala, Apache Spark, Apache Airflow, Python, AWS Kinesis, EMR, DynamoDB, SQS, S3, Kubernetes, Datadog

- **Founding engineer**: helped set the architecture, code conventions and CI/CD from day one for three services that take sensor data from real-time ingestion to analytics-ready datasets.
- **Real-time ingestion (Go)**: wrote a custom **AWS Kinesis consumer** for Sparkplug B sensor telemetry (birth and data messages) with DynamoDB checkpointing, consumer groups and shard handling, plus SQS integration.
- **Platform API (Go)**: an auth layer with JWT access and refresh tokens, RBAC, Google OAuth, email verification and password recovery; device, room, facility and manual-log domains; a reusable retry package and DB migrations, with tests across the controller, manager and repository layers.
- **Batch processing (Scala / Spark on EMR)**: jobs following the medallion architecture for ingestion, gold-layer aggregations, master dataset and dedup, product and metric runners, customer data egress and data migration, plus a carbon-registry integration and shared Spark, JDBC and S3-delivery utilities, each with tests.
- **Orchestration (Airflow)**: config-driven hourly and daily DAGs that launch the Spark jobs on EMR, backed by a config model for products (15-minute and 1-hour aggregations), gold layers, customers and providers, and a tested EMR client.
- **Operations**: Datadog metrics, traces and profiling, and Kubernetes manifests for dev and prod.

YouNet Media | Software Engineer **Mar 2022 – Mar 2025**
Social listening and e-commerce analytics

- **EcomHeat** (cross-channel e-commerce analytics): APIs that export 1M+ rows to Google Sheets, cutting manual work by 90%; price-history monitoring with daily alerts and SKU management; Airflow and BigQuery jobs for brand metrics; a NestJS monorepo and Highcharts dashboards.
- **AppCore**: designed core NestJS packages (Database, Cache, Redis and Config modules) published to npm and used across 3+ projects; caching cut load times by ∼40%.
- **Lucene Search Service**: pre-filtered records with Apache Lucene before indexing into Solr, reducing query load.

QKIT Software | Backend Developer **Jan 2021 – Dec 2022**
Outsourcing software company

- **Military 7A Bidding**: designed the PostgreSQL schema and backend from scratch for tracking medical-supply bidding packages, with JWT authentication, role-based access and Swagger API docs.
- **QKIT Admin**: CMS for website content, blogs, clients and recruitment (NestJS, PostgreSQL, Prisma).

Skills

Languages: TypeScript, JavaScript, Kotlin, Java, Go, Scala, Python, SQL, C++
Backend: Node.js, NestJS, Micronaut, Spring Boot, tRPC, REST, GraphQL, RabbitMQ
Data: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Apache Spark, Kafka, Airflow, Delta Lake
Frontend: React, TanStack Query and Form, zod, Tailwind CSS, Ant Design, Astro
Cloud & ops: AWS (SQS, Kinesis, EMR, S3, SageMaker), Docker, Kubernetes, Terraform, Datadog, GitHub Actions
Practices: transactional outbox, idempotent consumers, circuit breakers, rate limiting, feature-flagged releases, stacked PRs, AI-assisted development

Education

MSc in Information Systems | University of Information Technology, VNU-HCM **Dec 2024 – Present**
Thesis: a real-time mental-health state analysis system on Vietnamese social media (Kafka, Spark Streaming, Delta Lake, SageMaker); best model XLM-R at macro-F1 0.783.

BSc in Computer Science | University of Information Technology, VNU-HCM **Aug 2019 – Jun 2023**
GPA 8.5/10.

Projects & Certificates

karhdo.dev: personal blog and portfolio: Astro, Bun, Tailwind CSS, Drizzle on Neon Postgres, Vercel; this resume is built from LaTeX in the same repo.
Karhdo's Core: backend server packages for NestJS, published on npm.
TOEIC: 750/990.